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JSON-RPC API

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On this page 

In order for a software application to interact with the [Ethereum](#) blockchain - either by reading blockchain data or sending transactions to the network - it must connect to an Ethereum node.

For this purpose, every [Ethereum client](#) implements a [JSON-RPC specification](#) , so there is a uniform set of methods that applications can rely on regardless of the specific node or client implementation.

[JSON-RPC](#)  is a stateless, light-weight remote procedure call (RPC) protocol. It defines several data structures and the rules around their processing. It is transport agnostic in that the concepts can be used within the same process, over sockets, over HTTP, or in many various message passing environments. It uses JSON (RFC 4627) as data format.

Client implementations

Ethereum clients each may utilize different programming languages when implementing the JSON-RPC specification. See individual [client documentation](#) for further details related to specific programming languages. We recommend checking the documentation of each client for the latest API support information.

Convenience Libraries

While you may choose to interact directly with Ethereum clients via the JSON-RPC API, there are often easier options for dapp developers. Many [JavaScript](#) and [backend API](#) libraries exist to provide wrappers on top of the JSON-RPC API. With these libraries, developers can write intuitive, one-line methods in the programming language of their choice to initialize JSON-RPC requests (under the hood) that interact with Ethereum.

Consensus client APIs

This page deals mainly with the JSON-RPC API used by Ethereum execution clients. However, consensus clients also have an RPC API that allows users to query information about the node, request Beacon blocks, Beacon state, and other consensus-related information directly from a node. This API is documented on the [Beacon API webpage](#) [↗](#).

An internal API is also used for inter-client communication within a node - that is, it enables the consensus client and execution client to swap data. This is called the 'Engine API' and the specs are available on [GitHub](#) [↗](#).

Execution client spec

[Read the full JSON-RPC API spec on GitHub](#) [↗](#). This API is documented on the [Execution API webpage](#) [↗](#) and includes an Inspector to try out all the available methods.

Conventions

Hex value encoding

Two key data types get passed over JSON: unformatted byte arrays and quantities. Both are passed with a hex encoding but with different requirements for formatting.

Quantities

When encoding quantities (integers, numbers): encode as hex, prefix with "0x", the most compact representation (slight exception: zero should be represented as "0x0").

Here are some examples:

- 0x41 (65 in decimal)
- 0x400 (1024 in decimal)
- WRONG: 0x (should always have at least one digit - zero is "0x0")
- WRONG: 0x0400 (no leading zeroes allowed)
- WRONG: ff (must be prefixed 0x)

Unformatted data

When encoding unformatted data (byte arrays, account addresses, hashes, bytecode arrays): encode as hex, prefix with "0x", two hex digits per byte.

Here are some examples:

- 0x41 (size 1, "A")
- 0x004200 (size 3, "0B0")
- 0x (size 0, "")
- WRONG: 0xf0f0f (must be even number of digits)
- WRONG: 004200 (must be prefixed 0x)

The block parameter

The following methods have a block parameter:

- [eth_getBalance](#)
- [eth_getCode](#)
- [eth_getTransactionCount](#)
- [eth_getStorageAt](#)
- [eth_call](#)

When requests are made that query the state of Ethereum, the provided block parameter determines the height of the block.

The following options are possible for the block parameter:

- HEX String - an integer block number
- String "earliest" for the earliest/genesis block
- String "latest" - for the latest proposed block
- String "safe" - for the latest safe head block
- String "finalized" - for the latest finalized block
- String "pending" - for the pending state/transactions

Examples

On this page we provide examples of how to use individual JSON_RPC API endpoints using the command line tool, [curl](#) . These individual endpoint examples are found below in the [Curl examples](#) section. Further down the page, we also provide an [end-to-end example](#) for compiling and deploying a smart contract using a Geth node, the JSON_RPC API and curl.

Curl examples

Examples of using the JSON_RPC API by making [curl](#)  requests to an Ethereum node are provided below. Each example includes a description of the specific endpoint, its parameters, return type, and a worked example of how it should be used.

The curl requests might return an error message relating to the content type. This is because the `--data` option sets the content type to `application/x-www-form-urlencoded`. If your node does complain about this, manually set the header by placing `-H "Content-Type: application/json"` at the start of the call. The examples also do not include the URL/IP & port combination which must be the last argument given to curl (e.g., `127.0.0.1:8545`). A complete curl request including these additional data takes the following form:

SHELL

```
curl -H "Content-Type: application/json" -X POST --data '{"jsonrpc":"2.0","method":"w
```

Gossip, State, History

A handful of core JSON-RPC methods require data from the Ethereum network, and fall neatly into three main categories: *Gossip*, *State*, and *History*. Use the links in these sections to jump to each method, or use the table of contents to explore the whole list of methods.

Gossip Methods

These methods track the head of the chain. This is how transactions make their way around the network, find their way into blocks, and how clients find out about new blocks.

- [eth_blockNumber](#)
- [eth_sendRawTransaction](#)

State Methods

Methods that report the current state of all the data stored. The "state" is like one big shared piece of RAM, and includes account balances, contract data, and gas estimations.

- [eth_getBalance](#)
- [eth_getStorageAt](#)
- [eth_getTransactionCount](#)
- [eth_getCode](#)
- [eth_call](#)
- [eth_estimateGas](#)

History Methods

Fetches historical records of every block back to genesis. This is like one large append-only file, and includes all block headers, block bodies, uncle blocks, and transaction receipts.

- [eth_getBlockTransactionCountByHash](#)
- [eth_getBlockTransactionCountByNumber](#)
- [eth_getUncleCountByBlockHash](#)
- [eth_getUncleCountByBlockNumber](#)
- [eth_getBlockByHash](#)
- [eth_getBlockByNumber](#)
- [eth_getTransactionByHash](#)
- [eth_getTransactionByBlockHashAndIndex](#)
- [eth_getTransactionByBlockNumberAndIndex](#)
- [eth_getTransactionReceipt](#)
- [eth_getUncleByBlockHashAndIndex](#)
- [eth_getUncleByBlockNumberAndIndex](#)

JSON-RPC API Playground

You can use the [playground tool](#) [↗](#) to discover and try out the API methods. It also shows you which methods and networks are supported by various node providers.

JSON-RPC API Methods

web3_clientVersion

Returns the current client version.

Parameters

None

Returns

String - The current client version

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"web3_clientVersion","params":[],
3 // Result
4 {
5   "id":67,
6   "jsonrpc":"2.0",
7   "result": "Geth/v1.12.1-stable/linux-amd64/go1.19.1"
8 }
```

web3_sha3

Returns Keccak-256 (*not* the standardized SHA3-256) of the given data.

Parameters

1. DATA - The data to convert into a SHA3 hash

JS

```
1 params: ["0x68656c6c6f20776f726c64"]
```

Returns

DATA - The SHA3 result of the given string.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"web3_sha3","params":["0x68656c6c
3 // Result
4 {
```

```
5 | "id":64,  
6 | "jsonrpc": "2.0",  
7 | "result": "0x47173285a8d7341e5e972fc677286384f802f8ef42a5ec5f03bbfa254cb01fad"  
8 | }
```

net_version

Returns the current network id.

Parameters

None

Returns

String - The current network id.

The full list of current network IDs is available at chainlist.org [↗](#). Some common ones are:

- 1: Ethereum Mainnet
- 11155111: Sepolia testnet
- 560048 : Hoodi Testnet

Example

```
1 | // Request  
2 | curl -X POST --data '{"jsonrpc":"2.0","method":"net_version","params":[],"id":67}'  
3 | // Result  
4 | {  
5 |   "id":67,  
6 |   "jsonrpc": "2.0",  
7 |   "result": "3"  
8 | }
```

JS

net_listening

Returns true if client is actively listening for network connections.

Parameters

None

Returns

Boolean - true when listening, otherwise false.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"net_listening","params":[],"id":
3 // Result
4 {
5   "id":67,
6   "jsonrpc":"2.0",
7   "result":true
8 }
```

net_peerCount

Returns number of peers currently connected to the client.

Parameters

None

Returns

QUANTITY - integer of the number of connected peers.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"net_peerCount","params":[],"id":
3 // Result
4 {
5   "id":74,
6   "jsonrpc": "2.0",
```

```
7   "result": "0x2" // 2
8   }
```

eth_protocolVersion

Returns the current Ethereum protocol version. Note that this method is [not available in Geth](#) ↗.

Parameters

None

Returns

String - The current Ethereum protocol version

Example

```
1  // Request
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_protocolVersion","params":[]}'
3  // Result
4  {
5    "id":67,
6    "jsonrpc": "2.0",
7    "result": "54"
8  }
```

JS

eth_syncing

Returns an object with data about the sync status or false.

[Try endpoint in playground](#) ↗

Parameters

None

Returns

The precise return data varies between client implementations. All clients return `False` when the node is not syncing, and all clients return the following fields.

Object | Boolean, An object with sync status data or `FALSE`, when not syncing:

- `startingBlock`: QUANTITY - The block at which the import started (will only be reset, after the sync reached his head)
- `currentBlock`: QUANTITY - The current block, same as `eth_blockNumber`
- `highestBlock`: QUANTITY - The estimated highest block

However, the individual clients may also provide additional data. For example Geth returns the following:

```

1  {
2    "jsonrpc": "2.0",
3    "id": 1,
4    "result": {
5      "currentBlock": "0x3cf522",
6      "healedBytecodeBytes": "0x0",
7      "healedBytecodes": "0x0",
8      "healedTrieNodes": "0x0",
9      "healingBytecode": "0x0",
10     "healingTrieNodes": "0x0"

```

JSON

Show all (20) ▾

Whereas Besu returns:

```

1  {
2    "jsonrpc": "2.0",
3    "id": 51,
4    "result": {
5      "startingBlock": "0x0",
6      "currentBlock": "0x1518",
7      "highestBlock": "0x9567a3",
8      "pulledStates": "0x203ca",
9      "knownStates": "0x200636"

```

JSON

Show all (11) ▾

Refer to the documentation for your specific client for more details.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_syncing","params":[],"id":
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": {
8     startingBlock: '0x384',
9     currentBlock: '0x386',
```

Show all (18) ▾

eth_coinbase

Returns the client coinbase address.

[Try endpoint in playground](#) [↗](#)

Note: This method has been deprecated as of **v1.14.0** and is no longer supported. Attempting to use this method will result in a "Method not supported" error.

Parameters

None

Returns

DATA, 20 bytes - the current coinbase address.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_coinbase","params":[],"id":6
3 // Result
4 {
```

```
5   "id":64,  
6   "jsonrpc": "2.0",  
7   "result": "0x407d73d8a49eeb85d32cf465507dd71d507100c1"  
8 }
```

eth_chainId

Returns the chain ID used for signing replay-protected transactions.

[Try endpoint in playground](#) 

Parameters

None

Returns

chainId, hexadecimal value as a string representing the integer of the current chain id.

Example

```
1 // Request  
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_chainId","params":[],"id":67}'  
3 // Result  
4 {  
5   "id":67,  
6   "jsonrpc": "2.0",  
7   "result": "0x1"  
8 }
```

JS

eth_mining

Returns true if client is actively mining new blocks. This can only return true for proof-of-work networks and may not be available in some clients since [The Merge](#).

[Try endpoint in playground](#) 

Parameters

None

Returns

Boolean - returns true if the client is mining, otherwise false.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_mining","params":[],"id":71}'
3 //
4 {
5   "id":71,
6   "jsonrpc": "2.0",
7   "result": true
8 }
```

eth_hashrate

Returns the number of hashes per second that the node is mining with. This can only return true for proof-of-work networks and may not be available in some clients since [The Merge](#).

Try endpoint in playground [↗](#)

Parameters

None

Returns

QUANTITY - number of hashes per second.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_hashrate","params":[],"id":71}'
3 // Result
4 {
5   "id":71,
```

```
6   "jsonrpc": "2.0",  
7   "result": "0x38a"  
8   }
```

eth_gasPrice

Returns an estimate of the current price per gas in wei. For example, the Besu client examines the last 100 blocks and returns the median gas unit price by default

[Try endpoint in playground](#) 

Parameters

None

Returns

QUANTITY - integer of the current gas price in wei.

Example

```
1  // Request  
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_gasPrice","params":[],"id":7  
3  // Result  
4  {  
5    "id":73,  
6    "jsonrpc": "2.0",  
7    "result": "0x1dfd14000" // 8049999872 Wei  
8  }
```

JS

eth_accounts

Returns a list of addresses owned by client.

[Try endpoint in playground](#) 

Parameters

None

Returns

Array of DATA, 20 Bytes - addresses owned by the client.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_accounts","params":[],"id":1}'
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": ["0x407d73d8a49eeb85d32cf465507dd71d507100c1"]
8 }
```

eth_blockNumber

Returns the number of the most recent block.

[Try endpoint in playground](#) 

Parameters

None

Returns

QUANTITY - integer of the current block number the client is on.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_blockNumber","params":[],"id":1}'
3 // Result
4 {
5   "id":83,
6   "jsonrpc": "2.0",
7   "result": "0x4b7" // 1207
8 }
```

eth_getBalance

Returns the balance of the account at a given address.

[Try endpoint in playground](#) 

Parameters

1. DATA, 20 Bytes - address to check for balance.
2. QUANTITY|TAG - integer block number, or the string "latest", "earliest", "pending", "safe", or "finalized", see the [block parameter](#)

```
1 | params: ["0x407d73d8a49eeb85d32cf465507dd71d507100c1", "latest"]
```

JS

Returns

QUANTITY - integer of the current balance in wei.

Example

```
1 | // Request
2 | curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getBalance","params":["0x407
3 | // Result
4 | {
5 |   "id":1,
6 |   "jsonrpc": "2.0",
7 |   "result": "0x0234c8a3397aab58" // 158972490234375000
8 | }
```

JS

eth_getStorageAt

Returns the value from a storage position at a given address.

[Try endpoint in playground](#) 

Parameters

1. DATA, 20 Bytes - address of the storage.
2. QUANTITY - integer of the position in the storage.
3. QUANTITY|TAG - integer block number, or the string "latest", "earliest", "pending", "safe", "finalized", see the [block parameter](#)

Returns

DATA - the value at this storage position.

Example Calculating the correct position depends on the storage to retrieve. Consider the following contract deployed at 0x295a70b2de5e3953354a6a8344e616ed314d7251 by address 0x391694e7e0b0cce554cb130d723a9d27458f9298.

```

1  contract Storage {
2      uint pos0;
3      mapping(address => uint) pos1;
4      constructor() {
5          pos0 = 1234;
6          pos1[msg.sender] = 5678;
7      }
8  }
```

Retrieving the value of pos0 is straightforward:

JS

```

1  curl -X POST --data '{"jsonrpc":"2.0", "method": "eth_getStorageAt", "params": [
2      {"jsonrpc":"2.0", "id":1, "result": "0x0000000000000000000000000000000000000000000000000000000000000000"}

```

Retrieving an element of the map is harder. The position of an element in the map is calculated with:

JS

```

1  keccak(LeftPad32(key, 0), LeftPad32(map position, 0))

```


2. QUANTITY|TAG - integer block number, or the string "latest", "earliest", "pending", "safe" or "finalized", see the [block parameter](#)

JS

```
1 | params: [  
2 |     "0x407d73d8a49eeb85d32cf465507dd71d507100c1",  
3 |     "latest", // state at the latest block  
4 | ]
```

Returns

QUANTITY - integer of the number of transactions sent from this address.

Example

JS

```
1 | // Request  
2 | curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getTransactionCount","params"  
3 | // Result  
4 | {  
5 |     "id":1,  
6 |     "jsonrpc": "2.0",  
7 |     "result": "0x1" // 1  
8 | }
```

eth_getBlockTransactionCountByHash

Returns the number of transactions in a block from a block matching the given block hash.

[Try endpoint in playground](#) [↗](#)

Parameters

1. DATA, 32 Bytes - hash of a block

JS

```
1 | params: [ "0xd03ededb7415d22ae8bac30f96b2d1de83119632693b963642318d87d1bece5b" ]
```

Returns

QUANTITY - integer of the number of transactions in this block.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getBlockTransactionCountByHa
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x8b" // 139
8 }
```

JS

eth_getBlockTransactionCountByNumber

Returns the number of transactions in a block matching the given block number.

[Try endpoint in playground](#) [↗](#)

Parameters

1. QUANTITY|TAG - integer of a block number, or the string "earliest", "latest", "pending", "safe" or "finalized", as in the [block parameter](#).

```
1 params: [
2   "0x13738ca", // 20396234
3 ]
```

JS

Returns

QUANTITY - integer of the number of transactions in this block.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getBlockTransactionCountByNu
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x8b" // 139
8 }
```

eth_getUncleCountByBlockHash

Returns the number of uncles in a block from a block matching the given block hash.

[Try endpoint in playground](#) 

Parameters

1. DATA, 32 Bytes - hash of a block

JS

```
1 params: [ "0x1d59ff54b1eb26b013ce3cb5fc9dab3705b415a67127a003c3e61eb445bb8df2" ]
```

Returns

QUANTITY - integer of the number of uncles in this block.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getUncleCountByBlockHash","p
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x1" // 1
8 }
```

eth_getUncleCountByBlockNumber

Returns the number of uncles in a block from a block matching the given block number.

[Try endpoint in playground](#) 

Parameters

1. QUANTITY|TAG - integer of a block number, or the string "latest", "earliest", "pending", "safe" or "finalized", see the [block parameter](#)

```
1  params: [  
2    "0xe8", // 232  
3  ]
```

JS

Returns

QUANTITY - integer of the number of uncles in this block.

Example

```
1  // Request  
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getUncleCountByBlockNumber",  
3  // Result  
4  {  
5    "id":1,  
6    "jsonrpc": "2.0",  
7    "result": "0x0" // 0  
8  }
```

JS

eth_getCode

Returns code at a given address.

[Try endpoint in playground](#) 

Parameters

1. DATA, 20 Bytes - address
2. DATA, N Bytes - message to sign

Returns

DATA: Signature

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_sign","params":["0x9b2055d37
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0xa3f20717a250c2b0b729b7e5becbfff67fdaef7e0699da4de7ca5895b02a170a12
8 }
```

eth_signTransaction

Signs a transaction that can be submitted to the network at a later time using with [eth_sendRawTransaction](#).

Parameters

1. Object - The transaction object

- type:
- from: DATA, 20 Bytes - The address the transaction is sent from.
- to: DATA, 20 Bytes - (optional when creating new contract) The address the transaction is directed to.
- gas: QUANTITY - (optional, default: 90000) Integer of the gas provided for the transaction execution. It will return unused gas.
- gasPrice: QUANTITY - (optional, default: To-Be-Determined) Integer of the gasPrice used for each paid gas, in Wei.
- value: QUANTITY - (optional) Integer of the value sent with this transaction, in Wei.

- **data:** DATA - The compiled code of a contract OR the hash of the invoked method signature and encoded parameters.
- **nonce:** QUANTITY - (optional) Integer of a nonce. This allows to overwrite your own pending transactions that use the same nonce.

Returns

DATA, The RLP-encoded transaction object signed by the specified account.

Example

```
1 // Request
2 curl -X POST --data '{"id": 1,"jsonrpc": "2.0","method": "eth_signTransaction",'
3 // Result
4 {
5     "id": 1,
6     "jsonrpc": "2.0",
7     "result": "0xa3f20717a250c2b0b729b7e5becbffa67fdaef7e0699da4de7ca5895b02a170a
8 }
```

eth_sendTransaction

Creates new message call transaction or a contract creation, if the data field contains code, and signs it using the account specified in `from`.

Parameters

1. Object - The transaction object

- **from:** DATA, 20 Bytes - The address the transaction is sent from.
- **to:** DATA, 20 Bytes - (optional when creating new contract) The address the transaction is directed to.
- **gas:** QUANTITY - (optional, default: 90000) Integer of the gas provided for the transaction execution. It will return unused gas.
- **gasPrice:** QUANTITY - (optional, default: To-Be-Determined) Integer of the gasPrice used for each paid gas.

- **value:** QUANTITY - (optional) Integer of the value sent with this transaction.
- **input:** DATA - The compiled code of a contract OR the hash of the invoked method signature and encoded parameters.
- **nonce:** QUANTITY - (optional) Integer of a nonce. This allows to overwrite your own pending transactions that use the same nonce.

JS

```

1  params: [
2    {
3      from: "0xb60e8dd61c5d32be8058bb8eb970870f07233155",
4      to: "0xd46e8dd67c5d32be8058bb8eb970870f07244567",
5      gas: "0x76c0", // 30400
6      gasPrice: "0x9184e72a000", // 100000000000000
7      value: "0x9184e72a", // 2441406250
8      input:
9        "0xd46e8dd67c5d32be8d46e8dd67c5d32be8058bb8eb970870f072445675058bb8eb970

```

Show all (11) ▾

Returns

DATA, 32 Bytes - the transaction hash, or the zero hash if the transaction is not yet available.

Use [eth_getTransactionReceipt](#) to get the contract address, after the transaction was proposed in a block, when you created a contract.

Example

JS

```

1  // Request
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_sendTransaction","params":[{"id":1,
3  // Result
4  {
5    "id":1,
6    "jsonrpc": "2.0",
7    "result": "0xe670ec64341771606e55d6b4ca35a1a6b75ee3d5145a99d05921026d1527331"
8  }

```

eth_sendRawTransaction

Creates new message call transaction or a contract creation for signed transactions.

Parameters

1. DATA, The signed transaction data.

JS

```
1  params: [  
2    "0xd46e8dd67c5d32be8d46e8dd67c5d32be8058bb8eb970870f072445675058bb8eb970870f07  
3  ]
```

Returns

DATA, 32 Bytes - the transaction hash, or the zero hash if the transaction is not yet available.

Use [eth_getTransactionReceipt](#) to get the contract address, after the transaction was proposed in a block, when you created a contract.

Example

JS

```
1  // Request  
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_sendRawTransaction","params"  
3  // Result  
4  {  
5    "id":1,  
6    "jsonrpc": "2.0",  
7    "result": "0xe670ec64341771606e55d6b4ca35a1a6b75ee3d5145a99d05921026d1527331"  
8  }
```

eth_call

Executes a new message call immediately without creating a transaction on the blockchain. Often used for executing read-only smart contract functions, for example the `balanceOf` for an ERC-20 contract.

[Try endpoint in playground](#) 

Parameters

1. Object - The transaction call object

- **from:** DATA, 20 Bytes - (optional) The address the transaction is sent from.
- **to:** DATA, 20 Bytes - The address the transaction is directed to.
- **gas:** QUANTITY - (optional) Integer of the gas provided for the transaction execution. `eth_call` consumes zero gas, but this parameter may be needed by some executions.
- **gasPrice:** QUANTITY - (optional) Integer of the gasPrice used for each paid gas
- **value:** QUANTITY - (optional) Integer of the value sent with this transaction
- **input:** DATA - (optional) Hash of the method signature and encoded parameters. For details see [Ethereum Contract ABI in the Solidity documentation](#) .

2. QUANTITY|TAG - integer block number, or the string "latest", "earliest", "pending", "safe" or "finalized", see the [block parameter](#)

Returns

DATA - the return value of executed contract.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_call","params":[{"see above}]}'
3 // Result
4 {
5     "id":1,
6     "jsonrpc": "2.0",
7     "result": "0x"
8 }
```

JS

eth_estimateGas

Generates and returns an estimate of how much gas is necessary to allow the transaction to complete. The transaction will not be added to the blockchain. Note that the estimate may be significantly more than the amount of gas actually used by the transaction, for a variety of reasons including EVM mechanics and node performance.

[Try endpoint in playground](#) 

Parameters

See [eth_call](#) parameters, except that all properties are optional. If no gas limit is specified geth uses the block gas limit from the pending block as an upper bound. As a result the returned estimate might not be enough to execute the call/transaction when the amount of gas is higher than the pending block gas limit.

Returns

QUANTITY - the amount of gas used.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_estimateGas","params":[]}'
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x5208" // 21000
8 }
```

JS

eth_getBlockByHash

Returns information about a block by hash.

[Try endpoint in playground](#) 

Parameters

1. DATA, 32 Bytes - Hash of a block.

2. Boolean - If `true` it returns the full transaction objects, if `false` only the hashes of the transactions.

JS

```
1  params: [  
2    "0xdc0818cf78f21a8e70579cb46a43643f78291264dda342ae31049421c82d21ae",  
3    false,  
4  ]
```

Returns

Object - A block object, or `null` when no block was found:

- `number`: QUANTITY - the block number. `null` when its pending block.
- `hash`: DATA, 32 Bytes - hash of the block. `null` when its pending block.
- `parentHash`: DATA, 32 Bytes - hash of the parent block.
- `nonce`: DATA, 8 Bytes - hash of the generated proof-of-work. `null` when its pending block, `0x0` for proof-of-stake blocks (since The Merge)
- `sha3Uncles`: DATA, 32 Bytes - SHA3 of the uncles data in the block.
- `logsBloom`: DATA, 256 Bytes - the bloom filter for the logs of the block. `null` when its pending block.
- `transactionsRoot`: DATA, 32 Bytes - the root of the transaction trie of the block.
- `stateRoot`: DATA, 32 Bytes - the root of the final state trie of the block.
- `receiptsRoot`: DATA, 32 Bytes - the root of the receipts trie of the block.
- `miner`: DATA, 20 Bytes - the address of the beneficiary to whom the block rewards were given.
- `difficulty`: QUANTITY - integer of the difficulty for this block.
- `totalDifficulty`: QUANTITY - integer of the total difficulty of the chain until this block.
- `extraData`: DATA - the "extra data" field of this block.
- `size`: QUANTITY - integer the size of this block in bytes.
- `gasLimit`: QUANTITY - the maximum gas allowed in this block.

- gasUsed: QUANTITY - the total used gas by all transactions in this block.
- timestamp: QUANTITY - the unix timestamp for when the block was collated.
- transactions: Array - Array of transaction objects, or 32 Bytes transaction hashes depending on the last given parameter.
- uncles: Array - Array of uncle hashes.

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getBlockByHash","params":[
3 // Result
4 {
5     "jsonrpc": "2.0",
6     "id": 1,
7     "result": {
8         "difficulty": "0x4ea3f27bc",
9         "extraData": "0x476574682f4c5649562f76312e302e302f6c696e75782f676f312e342e"
```

Show all (31) ▾

eth_getBlockByNumber

Returns information about a block by block number.

[Try endpoint in playground](#) 

Parameters

1. QUANTITY|TAG - integer of a block number, or the string "earliest", "latest", "pending", "safe" or "finalized", as in the [block parameter](#).
2. Boolean - If true it returns the full transaction objects, if false only the hashes of the transactions.

```
1 params: [
2     "0x1b4", // 436
3     true,
4 ]
```

Returns See [eth_getBlockByHash](#)

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getBlockByNumber","params":['
```

Result see [eth_getBlockByHash](#)

eth_getTransactionByHash

Returns the information about a transaction requested by transaction hash.

[Try endpoint in playground](#) 

Parameters

1. DATA, 32 Bytes - hash of a transaction

JS

```
1 params: ["0x88df016429689c079f3b2f6ad39fa052532c56795b733da78a91ebe6a713944b"]
```

Returns

Object - A transaction object, or null when no transaction was found:

- blockHash: DATA, 32 Bytes - hash of the block where this transaction was in. null when its pending.
- blockNumber: QUANTITY - block number where this transaction was in. null when its pending.
- from: DATA, 20 Bytes - address of the sender.
- gas: QUANTITY - gas provided by the sender.
- gasPrice: QUANTITY - gas price provided by the sender in Wei.
- hash: DATA, 32 Bytes - hash of the transaction.

- **input:** DATA - the data send along with the transaction.
- **nonce:** QUANTITY - the number of transactions made by the sender prior to this one.
- **to:** DATA, 20 Bytes - address of the receiver. `null` when its a contract creation transaction.
- **transactionIndex:** QUANTITY - integer of the transactions index position in the block. `null` when its pending.
- **value:** QUANTITY - value transferred in Wei.
- **v:** QUANTITY - ECDSA recovery id
- **r:** QUANTITY - ECDSA signature r
- **s:** QUANTITY - ECDSA signature s

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getTransactionByHash","par
3 // Result
4 {
5   "jsonrpc":"2.0",
6   "id":1,
7   "result":{
8     "blockHash":"0x1d59ff54b1eb26b013ce3cb5fc9dab3705b415a67127a003c3e61eb445b
9     "blockNumber":"0x5daf3b", // 6139707
```

Show all (23) ▾

eth_getTransactionByBlockHashAndIndex

Returns information about a transaction by block hash and transaction index position.

[Try endpoint in playground](#) 

Parameters

1. DATA, 32 Bytes - hash of a block.
2. QUANTITY - integer of the transaction index position.

JS

```
1  params: [  
2    "0x1d59ff54b1eb26b013ce3cb5fc9dab3705b415a67127a003c3e61eb445bb8df2",  
3    "0x0", // 0  
4  ]
```

Returns See [eth_getTransactionByHash](#)

Example

JS

```
1  // Request  
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getTransactionByBlockHashAnd
```



Result see [eth_getTransactionByHash](#)

eth_getTransactionByBlockNumberAndIndex

Returns information about a transaction by block number and transaction index position.

[Try endpoint in playground](#) 

Parameters

1. QUANTITY|TAG - a block number, or the string "earliest", "latest", "pending", "safe" or "finalized", as in the [block parameter](#).
2. QUANTITY - the transaction index position.

JS

```
1  params: [  
2    "0x9c47cf", // 10241999  
3    "0x24", // 36  
4  ]
```

Returns See [eth_getTransactionByHash](#)

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getTransactionByBlockNumberA
```

Result see [eth_getTransactionByHash](#)

eth_getTransactionReceipt

Returns the receipt of a transaction by transaction hash.

Note That the receipt is not available for pending transactions.

Parameters

1. DATA, 32 Bytes - hash of a transaction

```
1 params: ["0x85d995eba9763907fdf35cd2034144dd9d53ce32cbec21349d4b12823c6860c5"]
```

Returns Object - A transaction receipt object, or null when no receipt was found:

- transactionHash : DATA, 32 Bytes - hash of the transaction.
- transactionIndex: QUANTITY - integer of the transactions index position in the block.
- blockHash: DATA, 32 Bytes - hash of the block where this transaction was in.
- blockNumber: QUANTITY - block number where this transaction was in.
- from: DATA, 20 Bytes - address of the sender.
- to: DATA, 20 Bytes - address of the receiver. null when its a contract creation transaction.
- cumulativeGasUsed : QUANTITY - The total amount of gas used when this transaction was executed in the block.
- effectiveGasPrice : QUANTITY - The sum of the base fee and tip paid per unit of gas.

- **gasUsed** : QUANTITY - The amount of gas used by this specific transaction alone.
- **contractAddress** : DATA, 20 Bytes - The contract address created, if the transaction was a contract creation, otherwise null.
- **logs**: Array - Array of log objects, which this transaction generated.
- **logsBloom**: DATA, 256 Bytes - Bloom filter for light clients to quickly retrieve related logs.
- **type**: QUANTITY - integer of the transaction type, 0x0 for legacy transactions, 0x1 for access list types, 0x2 for dynamic fees.

It also returns *either* :

- **root** : DATA 32 bytes of post-transaction state root (pre Byzantium)
- **status**: QUANTITY either 1 (success) or 0 (failure)

Example

```

1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getTransactionReceipt","pa
3 // Result
4 {
5   "jsonrpc": "2.0",
6   "id": 1,
7   "result": {
8     "blockHash":
9     "0xa957d47df264a31badc3ae823e10ac1d444b098d9b73d204c40426e57f47e8c3",

```

Show all (27) ▾

eth_getUncleByBlockHashAndIndex

Returns information about an uncle of a block by hash and uncle index position.

[Try endpoint in playground](#) 

Parameters

1. DATA, 32 Bytes - The hash of a block.

2. QUANTITY - The uncle's index position.

JS

```
1 | params: [  
2 |     "0x1d59ff54b1eb26b013ce3cb5fc9dab3705b415a67127a003c3e61eb445bb8df2",  
3 |     "0x0", // 0  
4 | ]
```

Returns See [eth_getBlockByHash](#)

Example

JS

```
1 | // Request  
2 | curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getUncleByBlockHashAndIndex"
```

Result see [eth_getBlockByHash](#)

Note: An uncle doesn't contain individual transactions.

eth_getUncleByBlockNumberAndIndex

Returns information about an uncle of a block by number and uncle index position.

[Try endpoint in playground](#) 

Parameters

1. QUANTITY|TAG - a block number, or the string "earliest", "latest", "pending", "safe", "finalized", as in the [block parameter](#).
2. QUANTITY - the uncle's index position.

JS

```
1 | params: [  
2 |     "0x29c", // 668  
3 |     "0x0", // 0  
4 | ]
```

Returns See [eth_getBlockByHash](#)

Note: An uncle doesn't contain individual transactions.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getUncleByBlockNumberAndIndex","params":[12,1],"id":1}'
```

Result see [eth_getBlockByHash](#)

eth_newFilter

Creates a filter object, based on filter options, to notify when the state changes (logs). To check if the state has changed, call [eth_getFilterChanges](#).

A note on specifying topic filters: Topics are order-dependent. A transaction with a log with topics [A, B] will be matched by the following topic filters:

- [] "anything"
- [A] "A in first position (and anything after)"
- [null, B] "anything in first position AND B in second position (and anything after)"
- [A, B] "A in first position AND B in second position (and anything after)"
- [[A, B], [A, B]] "(A OR B) in first position AND (A OR B) in second position (and anything after)"

• Parameters

1. Object - The filter options:

- fromBlock: QUANTITY|TAG - (optional, default: "latest") Integer block number, or "latest" for the last proposed block, "safe" for the latest safe block, "finalized" for the latest finalized block, or "pending", "earliest" for transactions not yet in a block.

- **toBlock:** QUANTITY|TAG - (optional, default: "latest") Integer block number, or "latest" for the last proposed block, "safe" for the latest safe block, "finalized" for the latest finalized block, or "pending", "earliest" for transactions not yet in a block.
- **address:** DATA|Array, 20 Bytes - (optional) Contract address or a list of addresses from which logs should originate.
- **topics:** Array of DATA, - (optional) Array of 32 Bytes DATA topics. Topics are order-dependent. Each topic can also be an array of DATA with "or" options.

```

1  params: [
2    {
3      fromBlock: "0x1",
4      toBlock: "0x2",
5      address: "0x8888f1f195afa192cfee860698584c030f4c9db1",
6      topics: [
7        "0x00000000000000000000000000000000a94f5374fce5edbc8e2a8697c15331677e6ebf0b",
8        null,
9      ]

```

Show all (15) ▾

Returns QUANTITY - A filter id.

Example

```

1  // Request
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_newFilter","params":["topic
3  // Result
4  {
5    "id":1,
6    "jsonrpc": "2.0",
7    "result": "0x1" // 1
8  }

```

eth_newBlockFilter

Creates a filter in the node, to notify when a new block arrives. To check if the state has changed, call [eth_getFilterChanges](#).

Parameters None

Returns QUANTITY - A filter id.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_newBlockFilter","params":[]},
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x1" // 1
8 }
```

eth_newPendingTransactionFilter

Creates a filter in the node, to notify when new pending transactions arrive. To check if the state has changed, call [eth_getFilterChanges](#).

Parameters None

Returns QUANTITY - A filter id.

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_newPendingTransactionFilter"
3 // Result
4 {
5   "id":1,
6   "jsonrpc": "2.0",
7   "result": "0x1" // 1
8 }
```

eth_uninstallFilter

Uninstalls a filter with given id. Should always be called when watch is no longer needed. Additionally Filters timeout when they aren't requested with [eth_getFilterChanges](#) for a period of time.

Parameters

1. QUANTITY - The filter id.

```
1  params: [  
2    "0xb", // 11  
3  ]
```

JS

Returns Boolean - true if the filter was successfully uninstalled, otherwise false.

Example

```
1  // Request  
2  curl -X POST --data '{"jsonrpc":"2.0","method":"eth_uninstallFilter","params":["  
3  // Result  
4  {  
5    "id":1,  
6    "jsonrpc": "2.0",  
7    "result": true  
8  }
```

JS

eth_getFilterChanges

Polling method for a filter, which returns an array of logs which occurred since last poll.

Parameters

1. QUANTITY - the filter id.

```
1  params: [  
2    "0x16", // 22
```

JS

3 |]

Returns Array - Array of log objects, or an empty array if nothing has changed since last poll.

- For filters created with `eth_newBlockFilter` the return are block hashes (DATA, 32 Bytes), e.g., `["0x3454645634534..."]`.
- For filters created with `eth_newPendingTransactionFilter` the return are transaction hashes (DATA, 32 Bytes), e.g., `["0x6345343454645..."]`.
- For filters created with `eth_newFilter` logs are objects with following params:
 - `removed`: TAG - `true` when the log was removed, due to a chain reorganization. `false` if its a valid log.
 - `logIndex`: QUANTITY - integer of the log index position in the block. `null` when its pending log.
 - `transactionIndex`: QUANTITY - integer of the transactions index position log was created from. `null` when its pending log.
 - `transactionHash`: DATA, 32 Bytes - hash of the transactions this log was created from. `null` when its pending log.
 - `blockHash`: DATA, 32 Bytes - hash of the block where this log was in. `null` when its pending. `null` when its pending log.
 - `blockNumber`: QUANTITY - the block number where this log was in. `null` when its pending. `null` when its pending log.
 - `address`: DATA, 20 Bytes - address from which this log originated.
 - `data`: DATA - variable-length non-indexed log data. (In *solidity*: zero or more 32 Bytes non-indexed log arguments.)
 - `topics`: Array of DATA - Array of 0 to 4 32 Bytes DATA of indexed log arguments. (In *solidity*: The first topic is the *hash* of the signature of the event (e.g., `Deposit(address,bytes32,uint256)`), except you declared the event with the anonymous specifier.)

- **Example**

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getFilterChanges","params"
3 // Result
4 {
5   "id":1,
6   "jsonrpc":"2.0",
7   "result": [{
8     "logIndex": "0x1", // 1
9     "blockNumber": "0x1b4", // 436
```

Show all (19) ▾

eth_getFilterLogs

Returns an array of all logs matching filter with given id.

Parameters

1. QUANTITY - The filter id.

JS

```
1 params: [
2   "0x16", // 22
3 ]
```

Returns See [eth_getFilterChanges](#)

Example

JS

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getFilterLogs","params":["0x
```

Result see [eth_getFilterChanges](#)

eth_getLogs

Returns an array of all logs matching a given filter object.

Parameters

1. Object - The filter options:

- **fromBlock**: QUANTITY|TAG - (optional, default: "latest") Integer block number, or "latest" for the last proposed block, "safe" for the latest safe block, "finalized" for the latest finalized block, or "pending", "earliest" for transactions not yet in a block.
- **toBlock**: QUANTITY|TAG - (optional, default: "latest") Integer block number, or "latest" for the last proposed block, "safe" for the latest safe block, "finalized" for the latest finalized block, or "pending", "earliest" for transactions not yet in a block.
- **address**: DATA|Array, 20 Bytes - (optional) Contract address or a list of addresses from which logs should originate.
- **topics**: Array of DATA, - (optional) Array of 32 Bytes DATA topics. Topics are order-dependent. Each topic can also be an array of DATA with "or" options.
- **blockHash**: DATA, 32 Bytes - (optional, **future**) With the addition of EIP-234, blockHash will be a new filter option which restricts the logs returned to the single block with the 32-byte hash blockHash. Using blockHash is equivalent to fromBlock = toBlock = the block number with hash blockHash. If blockHash is present in the filter criteria, then neither fromBlock nor toBlock are allowed.

JS

```
1  params: [  
2    {  
3      topics: [  
4        "0x00000000000000000000000000000000a94f5374fce5edbc8e2a8697c15331677e6ebf0b",  
5      ],  
6    },  
7  ]
```

Returns See [eth_getFilterChanges](#)

Example

```
1 // Request
2 curl -X POST --data '{"jsonrpc":"2.0","method":"eth_getLogs","params":[{"topics"
```

Result see [eth_getFilterChanges](#)

Usage Example

Deploying a contract using JSON_RPC

This section includes a demonstration of how to deploy a contract using only the RPC interface. There are alternative routes to deploying contracts where this complexity is abstracted away—for example, using libraries built on top of the RPC interface such as [web3.js](#) and [web3.py](#). These abstractions are generally easier to understand and less error-prone, but it is still helpful to understand what is happening under the hood.

The following is a straightforward smart contract called Multiply7 that will be deployed using the JSON-RPC interface to an Ethereum node. This tutorial assumes the reader is already running a Geth node. More information on nodes and clients is available [here](#). Please refer to individual [client](#) documentation to see how to start the HTTP JSON-RPC for non-Geth clients. Most clients default to serving on localhost:8545.

```
1 contract Multiply7 {
2     event Print(uint);
3     function multiply(uint input) returns (uint) {
4         Print(input * 7);
5         return input * 7;
6     }
7 }
```

The first thing to do is make sure the HTTP RPC interface is enabled. This means we supply Geth with the `--http` flag on startup. In this example we use the Geth node on a private development chain. Using this approach we don't need ether on the real network.

SHELL

```
geth --http --dev console 2>>geth.log
```

This will start the HTTP RPC interface on `http://localhost:8545`.

We can verify that the interface is running by retrieving the coinbase address (by obtaining the first address from the array of accounts) and balance using [curl](#) [↗](#). Please note that data in these examples will differ on your local node. If you want to try these commands, replace the request params in the second curl request with the result returned from the first.

SHELL

```
curl --data '{"jsonrpc":"2.0","method":"eth_accounts","params":[], "id":1}' -H "Content-Type: application/json" --data '{"id":1,"jsonrpc":"2.0","result":["0x9b1d35635cc34752ca54713bb99d38614f63c955"]}'
```

```
curl --data '{"jsonrpc":"2.0","method":"eth_getBalance", "params": ["0x9b1d35635cc34752ca54713bb99d38614f63c955", "latest"]}'
```

Because numbers are hex encoded, the balance is returned in wei as a hex string. If we want to have the balance in ether as a number we can use `web3` from the Geth console.

JS

```
1 web3.fromWei("0x1639e49bba16280000", "ether")
2 // "410"
```

Now that there is some ether on our private development chain, we can deploy the contract. The first step is to compile the Multiply7 contract to byte code that can be sent to the EVM. To install `solc`, the Solidity compiler, follow the [Solidity documentation](#) [↗](#). (You might want to use an older `solc` release to match [the version of compiler used for our example](#) [↗](#).)

The next step is to compile the Multiply7 contract to byte code that can be sent to the EVM.

SHELL

```
echo 'pragma solidity ^0.4.16; contract Multiply7 { event Print(uint); function multi
```

```
===== <stdin>:Multiply7 =====
```

```
Binary:
```

```
6060604052341561000f57600080fd5b60eb8061001d6000396000f300606060405260043610603f57600
```

Now that we have the compiled code we need to determine how much gas it costs to

deploy it. The RPC interface has an `eth_estimateGas` method that will give us an estimate.

SHELL

```
curl --data '{"jsonrpc":"2.0","method": "eth_estimateGas", "params": [{"from": "0x9b1
{"jsonrpc":"2.0","id":5,"result":"0x1c31e"}'
```

And finally deploy the contract.

SHELL

```
curl --data '{"jsonrpc":"2.0","method": "eth_sendTransaction", "params": [{"from": "0
{"id":6,"jsonrpc":"2.0","result":"0xe1f3095770633ab2b18081658bad475439f6a08c902d09159
```

The transaction is accepted by the node and a transaction hash is returned. This hash can be used to track the transaction. The next step is to determine the address where our contract is deployed. Each executed transaction will create a receipt. This receipt contains various information about the transaction such as in which block the transaction was included and how much gas was used by the EVM. If a transaction creates a contract it will also contain the contract address. We can retrieve the receipt with the `eth_getTransactionReceipt` RPC method.

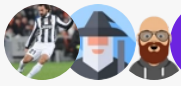
SHELL

```
curl --data '{"jsonrpc":"2.0","method": "eth_getTransactionReceipt", "params": ["0xe1
{"jsonrpc":"2.0","id":7,"result":{"blockHash":"0x77b1a4f6872b9066312de3744f60020cbd81
```

Our contract was created on `0x4d03d617d700cf81935d7f797f4e2ae719648262`. A null result instead of a receipt means the transaction has not been included in a block yet. Wait for a moment and check if your consensus client is running and retry it.

- [Execution clients](#)

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